

REVIEW

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Cardiovascular risk in children: a burden for future generations

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Abstract

Cardiovascular diseases are the main causes of death and health costs in developed countries. Although cardiovascular diseases are thought to affect only adulthood, the underlying process of atherosclerosis begins in the first decade of life. Epidemiological studies show that severity of atherosclerosis depends both on the number and intensity of risk factors. Early detection of cardiovascular risk in childhood is the most powerful tool to prevent cardiovascular accidents in adulthood and possibly reduce its consequent burden for the future.

A large amount of cardiovascular risk factors is already detectable in childhood and include non-modifiable elements, among which genetic factors and congenital heart diseases, and modifiable elements, which depend on environmental effects (e.g. lifestyle and nutrition). Regardless of the possibility to intervene on these factors, an early diagnosis is fundamental to ensure an optimal life expectancy in adulthood. The most important cardiovascular risk factors in the paediatric age and adolescence are excess weight, arterial hypertension, glucose metabolism and lipid metabolism alterations.

In this review we will discuss the main risk factors strictly correlated with cardiac and vessels diseases, focusing on their pathogenesis, diagnosis, and treatments.

Keywords: Congenital heart diseases, Congenital coronary anomalies, Kawasaki disease, Chronic kidney disease, Childhood cancer, Children

Introduction

Cardiovascular diseases (CVD) represent the main cause of death and health costs in industrialized countries. Several studies show that early atherosclerosis can be detected in childhood and its progression depends on exposure to several risk factors [1]. In fact, the initial vascular alterations of atherosclerotic pathology appear in the first decade of life [2]. Risk factors include metabolic alterations, hypertension, and chronic inflammatory conditions. Anyway, underlying structural or functional heart diseases are also important causes of CVD in childhood [1].

Some studies, such as the Bogalusa Heart Study and the Muscatine Study, have shown the association between exposure to various conventional cardiovascular risk factors in childhood and the development of subclinical atherosclerosis in old age. In particular, an increase in cholesterol and apolipoprotein levels in childhood is associated with an increase in carotid intima-media thickness (cIMT), a reduction in carotid elasticity and a compromised brachial endothelial function in adulthood [3].

A recent meta-analysis has recommended to perform a screening in children with cardiovascular risk factors from the end of the first decade of life, in order to detect subclinical atherosclerosis, because in children aged 9 years or older an increase in cIMT was predictive of the development of atherosclerosis in adulthood [4].

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